

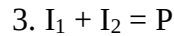
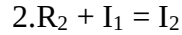
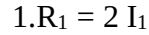
**Review of Chemical Kinetics

1. Rate of reaction schemes using De Donder Relations

Consider an arbitrary gas phase reaction



Which may take place according to following reaction mechanism



R_1, R_2 are reactants, P is the product and I_1, I_2 are reactive intermediates.

*De Donder (1927) related thermodynamic relationships to the chemical kinetics.

*The De Donder equation relates the rate of forward ($\vec{r}$) and backward ($\bar{r}$) reaction of elementary steps to the affinity of the reaction (A), as

$$\frac{\vec{r}}{\bar{r}} = \exp\left(\frac{A}{RT}\right) \quad (51)$$

Where the affinity (A) of chemical reaction is defined as the rate of change of free energy (G) w.r.t the extent of reaction (ξ)

$$A = -\left(\frac{\partial G}{\partial \xi}\right)_{T,P} \quad (52)$$

Free energy

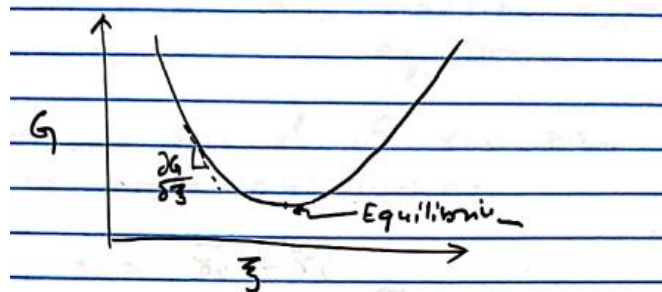
$$G = \sum_{j=1}^s N_j \bar{G}_j = \sum_{j=1}^s N_j \mu_j$$

Take derivative w.r.t ξ

$$\left(\frac{\partial G}{\partial \xi}\right)_{T,P} = \sum_{j=1}^s \frac{\partial N_j}{\partial \xi} \mu_j + \sum_{j=1}^s N_j \frac{\partial \mu_j}{\partial \xi} = \sum_{j=1}^s \alpha_j \mu_j \nu \quad (53)$$

$$A = - \sum_{j=1}^s \alpha_j G_j$$

At equilibrium, $A_j = 0$



*Chemical potential $\mu_j = \mu_{j,0} + RT \ln a_j$

Substitute in (54)

$$A_j = - \sum_j \alpha_{ij} [G_{j,0} + RT \ln a_j]$$

$$\exp\left(-\frac{A_i}{RT}\right) = \frac{\prod_j a_j^{\alpha_{ij}}}{K_i} \quad (55)$$

Where K_i is equilibrium constant

$$K_i = \exp\left(\frac{-\sum_j \alpha_{ij} G_{j,0}}{RT}\right)$$

Let $z_i = \exp\left(-\frac{A_i}{RT}\right) = \frac{\prod_j a_j^{\alpha_{ij}}}{K_i}$ (56) be the reversibility of step i or reaction i.

$z_i \rightarrow 0$ as reaction becomes irreversible. $z_i \rightarrow 1$ as reaction approaches equilibrium.

*The net rate of i^{th} reaction can be obtained as

$$r_i = \vec{r}_i - \hat{r}_i = \vec{r}_i \left(1 - \frac{\hat{r}_i}{\vec{r}_i}\right)$$

Use (51)

$$r_i = \vec{r}_i \left(1 - \exp\left(-\frac{A_i}{RT}\right)\right) = \vec{r}_i (1 - z_i) \quad (57)$$

This is a kinetic expression of elementary reaction using De Donder relation.

The reversibility z_i is for the 3-step reaction mechanism can be written as

$$z_1 = \frac{a_{I_1}^2}{K_1 a_{R_1}}$$

$$z_2 = \frac{a_{I_2}}{K_2 a_{R_2} a_{I_1}}$$

$$z_3 = \frac{a_p}{K_3 a_{I_1} a_{I_2}}$$

Note that the overall reversibility of the reaction is

$$z = \frac{a_p}{K a_{R_1} a_{R_2}} = z_1 z_2 z_3 \quad (59)$$

*Only two z_i 's are independent, the third is governed by overall z . This also implies overall affinity of the reaction $A = A_1 + A_2 + A_3$. In general,

$$A = \sum_{i=1}^m \sigma_i A_i \quad (60)$$

σ_i = number of times the i^{th} step must take place at each turnover of the overall reaction.

*The activities of intermediates can be written as

$$a_{I_1} = \sqrt{K_1 a_{R_1} z_1}$$

$$a_{I_2} = \sqrt{K_1 a_{R_1} z_1 K_2 a_{R_2} z_2}$$

Now the net rates of reactions can be written in terms of rate constant

$$r_1 = \vec{r}_1 (1 - z_1) = k_1 a_{R_1} (1 - z_1)$$

$$r_2 = \vec{r}_2(1 - z_2) = k a_{R_2} a_{I_1}(1 - z_2) = \sqrt{K_1} k_2 a_{R_1}^{\frac{1}{2}} a_{R_2} z_1^{\frac{1}{2}} (1 - z_2)$$

$$r_3 = \vec{r}_3(1 - z_3) = k_3 a_{I_1} a_{I_2}(1 - z_3) = K_1 K_2 k_3 a_{R_1} a_{R_2} z_1 z_2 \left(1 - \frac{z}{z_1 z_2}\right)$$

From stoichiometry, overall rxn = 1 + 2 + 3

$$\sigma_1 = \sigma_2 = \sigma_3 = 1$$

$$r = r_1 = r_2 = r_3$$

In general:

$$r = \sigma_i r_i \quad (63)$$

Equation 63 can also be obtained from pseudo-steady-state hypothesis (PSSH)

*The rate of change in concentration of I_1 and I_2 can be written as

$$\frac{dC_{I_1}}{dt} = 2r_1 - r_2 - r_3$$

$$\frac{dC_{I_2}}{dt} = r_2 - r_3$$

We know R_1 and R_2 decrease and P increases continuously as reaction proceeds. However, intermediate I_1 , I_2 do not change as much as R_1 , R_2 , P .

PSSH assumes $\frac{dC_{I_1}}{dt} = \frac{dC_{I_2}}{dt} = 0$. From which it follows $r_1 = r_2 = r_3$

$$\Rightarrow k_1 a_{R_1}(1 - z_1) = \sqrt{K_1} k_2 a_{R_1}^{\frac{1}{2}} a_{R_2} z_1^{\frac{1}{2}} (1 - z_2) = K_1 K_2 k_3 a_{R_1} a_{R_2} z_1 z_2 \left(1 - \frac{z}{z_1 z_2}\right) \quad (64)$$

Two equations and two unknown z_1, z_2 . Use (64) to determine z_1, z_2

*How many kinetic parameters are required to calculate rate from a reaction scheme?

Eg: We need three forward rate constants k_1, k_2, k_3 and three reverse rate constants or equilibrium constants K_1, K_2, K_3 .

There are total (2x m number of reactions) parameters required. Also $K = K_1 K_2 K_3 =$ known. There are (2m-1) unknown parameters in the rate expression.

*However, the minimum number of kinetic parameters required to compute rate for this 3-step scheme is 3.

$$C_1 = k_1 ; C_2 = \sqrt{K_1} k_2 ; C_3 = K_1 K_2 k_3 \quad (65)$$

Substitute (65) to (62),

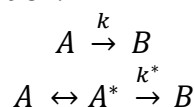
$$r_1 = C_1 a_{R_1}(1 - z_1)$$

$$r_2 = C_2 a_{R_1}^{1/2} a_{R_2} z_1^{1/2} (1 - z_2)$$

$$r_1 = C_3 a_{R_1} a_{R_2} z_1 z_2 \left(1 - \frac{z}{z_1 z_2}\right)$$

z_1, z_2 can be obtained from $r_1 = r_2 = r_3$,

Recall: transition state approximation:



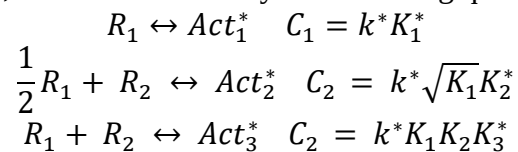
The reactant is equilibrated with the activated complex A^* . This is referred to as quasi equilibrium approx.

The rate constant

$$k = k^* K^* = k^* \exp \left[\frac{-\Delta G^*}{RT} \right]$$

Such that $r = k \cdot C_A$

The constants C_1, C_2, C_3 are controlled by the following quasi-equilibrium.



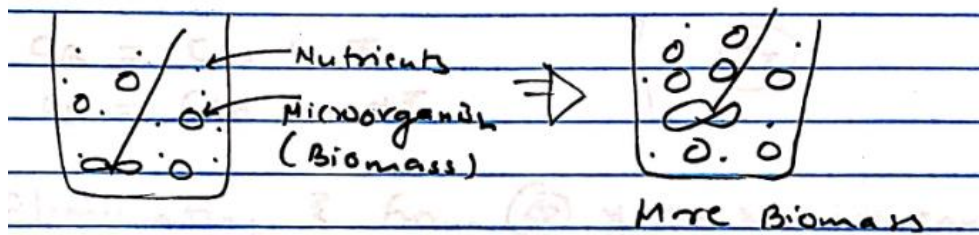
**Two commonly employed assumptions in the analysis of reaction mechanism are

1. Quasi-Equilibrium: Here we assume the faster reversible reaction at equilibrium and rest other reactions away from equilibrium.

2. Pseudo Steady state hypothesis (PSSH) : Here we assume the zero accumulation of intermediates at steady state.

2. Biochemical Reaction

Consider a well-stirred batch reactor in which biomass (of microorganism) is reacting with environmental nutrients to produce more biomass.



The process can be abstracted by



This growth reaction converts α mass units of substrate into a unit mass of fresh biomass.

The rate of reaction

$$r = \frac{\mu C_s C_B}{K_s + C_s}$$

Consider a constant volume well mixed batch reactor with initial concentrations of S and B as $C_{s,0}, C_{B,0}$ respectively. We are interested in determining $C_B(t), C_s(t)$.

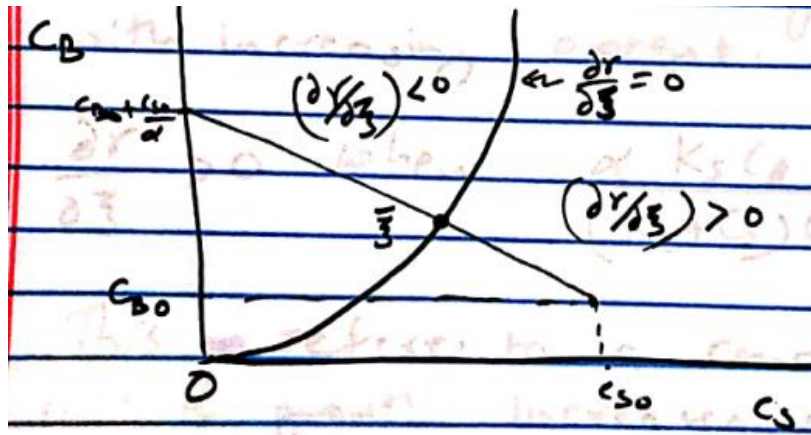
Let ξ be the extent of growth, so that $C_B = C_{B,0} + \xi$ and $C_s = C_{s,0} - \alpha \xi$

Eliminating ξ from (68), yield rxn invariant

$$\alpha C_B + C_s \leftrightarrow C_{s,0} + \alpha C_{B,0} \quad (69)$$

The max reaction extent can be realized when $C_s = 0$ from (68)

$$\xi_{max} = \frac{C_{S,0}}{\alpha} \quad (70)$$



Substitute (68) in (67)

$$r(\xi) = \frac{\mu(C_{S,0} - \alpha\xi)(C_{B,0} + \xi)}{K_s + C_{S,0} - \alpha\xi} \quad (71)$$

Differentiate w.r.t ξ

$$\begin{aligned} \frac{\partial r}{\partial \xi} &= \frac{\mu(C_{S,0} - \alpha\xi)}{K_s + C_{S,0} - \alpha\xi} - \alpha \frac{K_s \mu(C_{B,0} + \xi)}{K_s + C_{S,0} - \alpha\xi} = \frac{\mu C_s}{K_s + C_s} - \alpha \frac{K_s \mu C_B}{(K_s + C_s)^2} \\ &= \frac{\mu C_s}{K_s + C_s} \left[1 - \frac{\alpha K_s C_B}{C_s(K_s + C_s)} \right] \quad (72) \end{aligned}$$

$$\frac{\partial r}{\partial \xi} \leq 0 \text{ when } \frac{\alpha K_s C_B}{C_s(K_s + C_s)} \geq 1$$

This means the rate of growth decreases with increasing extent.

$$\frac{\partial r}{\partial \xi} > 0 \text{ when } \frac{\alpha K_s C_B}{C_s(K_s + C_s)} \leq 1$$

This refers to a condition when rate of growth increases with extent.

When $\frac{\partial r}{\partial \xi} = 0$, from (72)

$$C_B = \frac{1}{\alpha K_s} (K_s + C_s) C_s$$

Point of intersection of (69) and (73) represents critical value of ξ , below which the reaction is auto catalytic.

